

Research Topic	Citation	Abstract	Context	Research Questions or Hypotheses	Method	Method 1	Method 2	Analysis 1	Analysis 2	Results	Main Takeaway or Main Point	Impact and Importance (0-10)	Scientific Validity and Rigor (0-10)	Method Options	Analysis Options
Frustration in games	Ballou, N., & Detering, S. (2023). "I Just Wanted to Get It Over and Done With": A Grounded Theory of Psychological Need Frustration in Violent Games. <i>Proceedings of the ACM</i> , 16(1), 1-14.	Psychological need frustration experiences like failure, loneliness, or coercion—is emerging Cybersecurity is a pressing matter, and a lot of the	existing research on game-related need frustration relies on general instruments with unclear content evaluates the effectiveness of contextual and	RQ1: What kinds of need-frustrating experiences, if any, do people report in games? RQ1: Does game-based phishing training outperform	semi-structured interview	Interviews		Qualitative Data Analysis (Grounded Theory, Thematic Analysis)		(1) (unconscious) expectations for the kind and degree of autonomy, competence, and (1) Game-based training significantly improved detection	Expectations moderate need frustration in games interactive, contextual game-based training	4	7	6 Controlled Experiment with Random Assignment	Comparison of Means (t-test, ANOVA)
Game-Based Training for Phishing Detection	Kubliay, Elif, et al. "Can you spot a scam?" measuring and improving scam identification ability." SSRN Preprint, 2023.	The expansion of digital financial services leads to	responds to a gap in scam research: the lack of a	RQ1: Can scam identification ability be reliably	Survey-based experimental design	Survey Questionnaire Using Psychometric Likert		Qualitative Data Analysis (Grounded Theory, Thematic Analysis)		(1) Scam identification ability varies widely across	Scam detection is a quantifiable cognitive skill, and	7	7	8 Survey Questionnaire without Random Assignment	Regression
Scam Identification Ability	Roy, Abhishek, et al. "Shieldup: Inoculating Users against Online Scams Using a Game Based Intervention." <i>Proceedings of the ACM</i> , 16(1), 1-14.	Online scams are a growing threat in India, impacting	addresses the growing prevalence of online scams and	RQ1: Does playing the ShieldUp! game significantly improve	Participants assigned to game-based intervention	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Significant improvement in scam detection	Game-based psychological inoculation can have	8	7	7 Telemetry or Automated Collection of Behavioral Data	Structural Equation Modeling
Information Seeking About Phishing	Williams, Emma J, and Adam N Joinson. "Developing a measure of information seeking about	Phishing e-mails are fraudulent e-mails used to gain access	addresses the gap in understanding what	Do perceived threat severity, vulnerability, self-	Study 1: Developed item pool based on	Controlled Experiment with Random Assignment	Survey Questionnaire Using Psychometric Likert	Qualitative Data Analysis (Grounded Theory, Thematic Analysis)		(1) Response efficacy and self-efficacy predicted	Motivational factors (especially perceived efficacy)	7	7	7 Open-Ended Qualitative Questions	Lexical or Text Analysis
Phishing Email Suspicion Test	Hakim, Zaid M., et al. "The phishing email suspicion test (PEST) a lab-based task for	Phishing emails constitute a major problem, linked to	introduces the PEST, a lab-based measure designed	RQ1: Can PEST produce a valid measurement of	Participants rate a series of emails (phishing and	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) More effective real-world phishing emails were more	The PEST provides a quantifiable lab-based metric for	7	6	6 Interviews	Lexical or Text Analysis
Improving Phishing Reporting Using Security	Jensen, Matthew L., et al. "Improving phishing reporting using security gamification."	Phishing is an increasing threat that causes billions	explores how security gamification elements can	RQ1: Does incorporating gamification (e.g.,	Three experiments in a simulated workplace	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Public attribution with rewards/punishment	Security gamification can enhance phishing reporting	8	7	7 Focus Groups	
Game for Email Phishing Awareness Training	Misra, Gaurav, et al. "Phish Phinder: A Game Design Approach to Enhance User	Phishing threats are on the rise, especially through	phishing attacks are increasing 18n workplace email	RQ1: Can a serious game improve a user's ability to	Developed a serious game simulation for phishing scenarios	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Participants achieved 92.62% accuracy in	Serious games can significantly improve phishing detection	8	7	7 Diary Method	
Evaluating Phishing Awareness Strategies: A	Jayakrishnan, Gokul, et al. "PhickMail: A serious game for email phishing awareness	This paper investigates the effectiveness of	organizations commonly rely on awareness	RQ1: Which awareness strategy (traditional education	Participants tested for phishing susceptibility.	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Both education and game-based training improved	Phishing awareness programs are helpful but must be adapted	6	6	6 Experience Sampling Method	
Game Design Approach to Enhance User	Le-Nghe, Esperance Ngo, et al. "Evaluating phishing awareness strategies: A comparative study of	Phishing is an especially challenging cyber	many anti-phishing tools fail because they ignore the	RQ1: Can gamified learning improve user confidence in	Designed a gamified training platform with interactive	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Users gained conceptual and procedural	Gamified cybersecurity training can improve	7	6	6 Observation	
The roles of phishing knowledge, cue utilization, and	Sturman, Daniel, et al. "The roles of phishing knowledge, cue	This study investigated the roles of phishing	many users struggle to identify phishing emails even when	RQ1: Does phishing knowledge improve phishing detection	145 participants completed an email classification task	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Cue utilization was the strongest predictor of phishing	Training users to recognize specific phishing cues is	8	7	7 Usability Testing	
Cognitive elements of learning and discriminability in anti-phishing	Singh, Kuldeep, et al. "Cognitive elements of learning and discriminability in anti-phishing	People adjust decisions based on their experiences;	many anti-phishing training programs fail because they	RQ1: Does training feedback improve users' ability to	Laboratory experiment where participants were	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Detailed feedback significantly	Training programs should include frequent examples	7	8	8 Heuristic Evaluation	
Advancing Phishing Email Detection: A Comparative Study	Althwajry, Najwa, et al. "Advancing phishing email detection: A comparative study of deep	Phishing is one of the most dangerous attacks targeting	phishing emails are increasing rapidly, requiring more	RQ1: Which deep learning architectures	Researchers trained CNN-based deep learning models on	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Hybrid CNN + recurrent models improved phishing	Advanced deep learning models can significantly improve	6	7	7 Design Guidelines	
Simulated Phishing Attack and Embedded Training	Yeoh, William, et al. "Simulated phishing attack and embedded training campaign." <i>Journal of</i>	Phishing attacks are costly for both organizations and	phishing awareness programs are commonly used by	RQ1: How effective are simulated phishing campaigns	Case study of a large organizational phishing awareness	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Repeated phishing simulations helped users learn	Phishing awareness training works best when it includes	7	7	7 Task Performance	
Riskio! A Serious Game for Cyber Security Awareness	Hart, Stephen, et al. "Riskio! A serious game for cyber security awareness and Education."	Cyber attacks are increasing in number and sophistication,	traditional cybersecurity training can be	RQ1: Can serious games improve cybersecurity	Developed a tabletop cybersecurity game	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Players showed improved understanding of	Serious games can improve cybersecurity	6	6	6 Time on Task	
Phishing Awareness via Game-Based Learning	Rahartomo, Argianto, et al. "Phishing awareness via game-based learning." <i>2025 IEEE/ACM</i>	The increased use of digital devices and applications has	phishing attacks continue to grow with increased digital	RQ1: Can game-based learning increase phishing	Developed a serious game simulating various phishing	Controlled Experiment with Random Assignment		Comparison of Means (t-test, ANOVA)		(1) Players showed 24% improvement in phishing awareness.	Interactive cybersecurity games significantly improve	7	7	7 Scale Development	
														System Design	
														New Methods	
														Theories	
														Observation of Behavior	
														Ethics and policy implications	
														Literature Review	